Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Field Planting and Crop Establishment

1.1 Agro-Climatic Sowing Windows and Regional Calendar Timelines

- **Zone-Specific Planting Calendars across India:**
  - *North Zone (Punjab, Haryana, Rajasthan):* Irrigated cotton ecosystem (>97% irrigated area). Sowing commences April 1 and completes by May 15. Timely sowing within this 6-week window ensures optimal thermal units for vegetative growth and allows the crop canopy to escape early-season Cotton Leaf Curl Virus (CLCuD) outbreaks and pink bollworm peaks.
  - *Central Zone (Gujarat, Maharashtra, Madhya Pradesh):* Rainfed-dominated ecosystem (77% rainfed area). Sowing aligns with the arrival of the South-West monsoon, spanning June 15 to July 15. In irrigated pockets, pre-monsoon sowing is taken up in May to advance crop maturity.
  - *South Zone (Karnataka, Andhra Pradesh, Telangana, Tamil Nadu):* Multi-season ecosystem (60% rainfed area). Sowing spans June to August for Kharif crops, and August to September for late-season or winter irrigated crops in Tamil Nadu.
- **Thermal Requirements and Sowing Diurnal Hours:**
  - Optimal soil temperature for rapid cotton seed germination ranges between 20°C and 30°C at 5 cm soil depth.
  - Sowing operations during hot summer months (April–May in North India) must be conducted exclusively during early morning (0600 to 1000 h) and late evening hours (1600 to 1900 h) to prevent soil drying and thermal scorching of emerging seed embryos.
- **Hazards of Untimely Sowing:**
  - *Early Sowing (Prior to April 1):* Exposes germinating seedlings to severe atmospheric heat (> 42°C) and loose dry surface soil, causing extreme seedling burning, hypocotyl scorching, and poor plant stand.
  - *Late Sowing (After May 15 in North Zone / After July 15 in Central Zone):* Delays reproductive onset, exposes young tender foliage to peak whitefly vector populations and severe CLCuD infection, extends boll filling into cold winter months, and increases pink bollworm damage.
- **Refugia Strategy and Resistance Management for Bt Cotton:**

  - When planting Bt cotton hybrids expressing Cry1Ac and Cry2Ab endotoxins (Bollgard II), structured refuge planting is mandatory to delay bollworm resistance development.
  - *Structured Refuge Sowing:* Plant 5 border rows of non-Bt cotton around the outer perimeter of the Bt cotton field (or allocate 20% field area to non-Bt cotton under protected conditions / 5% under unprotected conditions).
  - *Refugia-in-Bag (RIB) Technology:* Modern commercial seeds are supplied as RIB formulations containing a 5% to 10% non-Bt seed mixture blended inside the Bt seed bag, eliminating the need for separate border row layout.

1.2 Cultivar Selection, Architectural Traits, and High-Density Planting Systems

- **Genotypic Species Classification:**
  - *Gossypium hirsutum (American Cotton / Narma):* Medium to extra-long staple length (25–32 mm), high yield potential, dominant in irrigated and commercial hybrid belts. Available as Bt hybrids, non-Bt hybrids, and open-pollinated varieties.
  - *Gossypium arboreum (Desi Cotton):* Indigenous species possessing inherent tolerance to sap-sucking pests (whiteflies, jassids), CLCuD, drought, high temperatures, and saline soils. Highly suited for low-input rainfed conditions and organic production.
  - *Gossypium barbadense (Extra-Long Staple Cotton):* High-value ELS cotton (>32 mm staple length) grown in localized pockets of South India.
  - *Gossypium herbaceum (Levant Cotton):* Drought-hardy Desi species grown in rainfed saline tracts of Gujarat and Karnataka.
- **Canopy Architectural Classification:**
  - *Compact Cultivars:* Short main stems, erect growth habit, short sympodial fruiting branches, zero to minimal monopodial vegetative branches, and synchronous boll maturity. Ideal for high-density close spacing.
  - *Semi-Spreading Cultivars:* Moderate height, 1–2 monopodial branches, medium sympodial length; suitable for standard irrigated spacings.
  - *Spreading Cultivars:* Tall, vigorous growth, multiple monopodial branches, long spreading sympodia; requires wide row spacing in fertile or rainfed soils.
- **High Density Planting System (HDPS):**
  - *Concept:* Agronomic technique utilizing compact, short-duration non-Bt or Bt varieties planted at high plant populations to maximize light interception, promote early boll setting, accelerate cutout, and enable mechanical single-pass picking.
  - *HDPS Geometry:* Sown at ultra-narrow spacing of 67.5 cm × 15 cm or 45 cm × 10 cm, establishing 98,765 to 2,22,222 plants per hectare (compared to standard hybrid densities of 15,432 to 24,691 plants/ha).

  - *Yield Advantage:* Produces 20–30% higher seed cotton yields in marginal or rainfed soils by increasing the total number of bolls per unit area rather than per plant.

- **Desi Cotton Diversification for Climate Resilience:**
  - Desi cotton varieties (e.g., CICR 1, CICR 3, HD 423, HD 432, FDK 124, RG 8) and Desi hybrids (e.g., CICR 2, AAH 1, FADH 9, RAJ.DH9) provide effective pest insurance in CLCuD and whitefly hotspot zones.
  - Shattering-tolerant Desi genotypes (such as variety CICR 3 and hybrid CICR 2) prevent seed cotton shedding during mature boll opening, reducing harvest labor costs.
  - Desi cotton cultivars must be sown before April 15 in North India to prevent early seedling heat burning and maintain uniform stands.

1.3 State-Wise Seed Rate Requirements and Plant Population Metrics

- **Seed Rate Guidelines by Species and State (per hectare basis):**
  - *American Bt Cotton Hybrids:* Standard seed rate is 2.375 to 2.970 kg/ha of Bt seed, supplemented with 250 to 500 g of non-Bt refugia seed (or packaged as 10.0 kg/ha total seeds under state recommended refuge protocols).
  - *American Non-Bt Cotton Varieties:* Recommended seed rate is 8.75 to 10.0 kg/ha across Haryana, Punjab, and Rajasthan.
  - *Desi Cotton Varieties (Desi Kapas):* Recommended seed rate is 7.5 kg/ha in Haryana and Punjab, and 11.25 to 15.0 kg/ha in Rajasthan.
  - *Desi Cotton Hybrids:* Standardized seed rate is 3.125 kg/ha across North Zone states.
  - *High Density Planting System (HDPS):* Requires 12.0 to 15.0 kg/ha of delinted seed due to closer plant-to-plant spacing.

- **Spacing Configurations and Plant Population Density Metrics:**
  - *Desi Cotton Varieties:* Sown at Row-to-Row (R×R) distance of 67.5 cm and Plant-to-Plant (P×P) distance of 30 cm in Haryana and Rajasthan (establishing 49,382 plants/ha), and 67.5 cm × 45 cm in Punjab (establishing 32,921 plants/ha).
  - *Desi Cotton Hybrids:* Sown at 67.5 cm × 60 cm spacing across Haryana, Punjab, and Rajasthan, establishing 24,691 plants/ha.
  - *American Cotton Varieties (Non-Bt Narma):* Sown at 67.5 cm × 30 cm in Haryana and Rajasthan (49,382 plants/ha), and 67.5 cm × 60 cm in Punjab (24,691 plants/ha).
  - *American Bt and Non-Bt Hybrids:* Sown at wide row geometry to accommodate vigorous canopy expansion. Spacing ranges from 67.5 cm × 75 cm in Punjab (19,753 plants/ha); 100 cm × 45 cm or 67.5 cm × 60 cm in Haryana (22,222 to 24,691 plants/ha); and 108 cm × 60 cm or 67.5 cm × 90 cm in Rajasthan (15,432 to 16,460 plants/ha).

  - *American Cotton HDPS (Narma):* Standardized spacing of 67.5 cm × 15 cm across Haryana, Punjab, and Rajasthan, establishing 98,765 plants/ha.

- **Mathematical Determination of Plant Population:** Total plant population per hectare is calculated using the formula: $PlantPopulation/ha = \frac{10{,}000m^2}{RowSpacing(m) \times PlantSpacing(m)}$. Maintaining target plant density ensures optimum light interception and maximum boll yield per unit land area.

1.4 Multi-Tier Seed Treatments and Acid Delinting Protocols

- **Commercial Acid Delinting Protocol:**
  - *Chemical Procedure:* Fuzzy cottonseed is treated with technical concentrated sulfuric acid (H_2SO_4, specific gravity 1.84) @ 100 mL per kg of seed in acid-resistant delinting drums.
  - *Reaction Mechanics:* The mixture is rotated continuously for 2 to 3 minutes until the surface fuzzy fibers dissolve completely, turning seeds dark brown/black.
  - *Washing and Neutralization:* Delinted seeds are immediately dumped into clean water drums and washed 3 to 4 times to strip residual acid. Seeds are neutralized by washing with a 1% lime water solution ($Ca(OH)_2$) or calcium carbonate, then spread under shade to dry to 8% moisture content.
  - *Phytosanitary Agronomic Benefits:* Destroys seed-borne bacterial blight spores (*Xanthomonas citri* pv. *malvacearum*), eliminates diapausing pink bollworm larvae inside double-joined seeds, removes germination-inhibiting fuzz, enables smooth flow through mechanical seed drills, and accelerates seed imbibition and germination by 2 to 3 days.

- **Sequential Seed Treatment Rule (F-I-B Order):**
  - Chemical and biological seed dressings must be applied in a strict sequential order: Fungicide first, Insecticide second, and Bioagent third ($F \rightarrow I \rightarrow B$).
  - Applying bioagents before chemical fungicides destroys beneficial microbial spores, rendering biological treatment ineffective.

- **Fungicidal Seed Treatment Formulations:**
  - Treat delinted seeds with Carboxin 37.5% + Thiram 37.5% DS @ 3.5 g/kg seed, or Thiram 75% WS @ 3.0 g/kg seed, or Carbendazim 50% WP @ 1.0–2.0 g/kg seed.
  - Advanced systemic combination: Sedaxane 2.5% + Fludioxonil 2.5% + Thiamethoxam 26.25% FS @ 4.0 mL/kg seed, providing multi-pathogen protection against *Rhizoctonia*, *Pythium*, and *Fusarium* root rots.

- **Systemic Insecticidal Seed Treatment Formulations:**
  - Treat seeds with Imidacloprid 70% WS @ 7.0 g/kg seed, or Imidacloprid 48% FS @ 5.0 to 9.0 mL/kg seed, or Thiamethoxam 30% FS @ 5.0 mL/kg seed, or Thiamethoxam 70% WS @ 4.0 to 5.0 g/kg seed.

  - Systemic insecticides translocate into emerging cotyledons and true leaves, protecting young plants against sucking pests (jassids, thrips, whiteflies, aphids) for the first 35 to 45 Days After Sowing (DAS) without requiring early foliar sprays.

- **Bactericidal and Bio-Agent Seed Dressings:**
  - *Bacterial Blight Protection:* Soak seeds in a 40 to 50 ppm solution of Streptomycin sulphate + Tetracycline hydrochloride (100 mg in 2 L water per kg seed) for 2 hours prior to drying and sowing.
  - *Biological Seed Protection:* Treat seeds with bio-control formulations of *Trichoderma viride* or *Trichoderma harzianum* (1% WP @ 5.0 to 10.0 g/kg seed) or *Pseudomonas fluorescens* (@ 10 g/kg seed) to enhance rhizosphere health and suppress soil-borne fungal pathogens.

1.5 Sowing Methods and Precision Field Machinery

- **Precision Tractor-Operated Planting Machinery:**
  - *Pneumatic Cotton Seed Planter-cum-Fertilizer Drill:* State-of-the-art planter utilizing tractor PTO-driven vacuum suction plates. Deliver single-seed singulation at exact hill distances without seed damage, maintaining precise seed depth (3–5 cm) and simultaneously drilling basal fertilizer in a side-band 7 cm away from seed rows.
  - *Inclined Plate Planter-cum-Fertilizer Drill:* Mechanical planter featuring rotating inclined cell-wheels tailored for acid-delinted seeds. Ensures uniform seed drop, prevents seed crowding, and lowers seed rate requirement by 20–30%.
  - *Walk-Behind Seed-cum-Fertilizer Drill:* Hand-guided mechanical planter suitable for smallholder farms where seeds are manually dropped into seed hoppers.
  - *Manually-Operated Mechanical Seed Dibblers:* Handheld push-dibblers featuring spring-loaded beak jaws that penetrate soil to a set depth, dropping individual seeds into moist soil.

- **Classification of Field Sowing Layouts:**
  - *Flat Bed Sowing:* Conventional planting on flat, level seedbeds using seed drills or manual dibbling. Best suited for well-drained loamy soils under canal or sprinkler irrigation.
  - *Narrow Raised Bed Sowing (Bed Top Planting):* Form narrow raised beds using a bed former. Seeds are dibbled on top of raised beds. Improves soil aeration, reduces surface crusting, and facilitates light furrow irrigation.
  - *Narrow Raised Bed Sowing (Furrow Planting):* Seeds are dibbled inside the furrows between raised beds. Ideal for dry regions or light sandy soils, placing seeds closer to subsoil moisture.
  - *Broad Raised Bed Sowing (One-Side or Both-Sides Planting):* Form broad raised beds (100–120 cm top width). Seeds are planted along one edge or both edges of the broad bed top. Saves 25–30% irrigation water and provides excellent drainage during heavy monsoon rains.

  - *Ridge and Furrow System:* Form ridges using a ridge maker. Seeds are dibbled on the side of ridges (middle-third height). Prevents direct water contact with seedling crowns during furrow irrigation.

- **Manual Dibbling Protocol:**
  - Executed after complete field preparation and attainment of proper *Wattar* soil moisture following pre-sowing *Rauni* irrigation.
  - Laborers walk along marked row lines, dibbling 1 to 2 delinted seeds per hill at precise 3 to 5 cm depth using handheld wooden sticks or iron dibblers, gently pressing soil over seeds.

1.6 Seedbed Hydraulics, Sowing Depth, and Soil Moisture (*Wattar*) Management

- **Seedbed Soil Moisture State (*Wattar*):**
  - Sowing must take place precisely when the seedbed attains the *Wattar* (workable moisture) condition following heavy pre-sowing *Rauni* irrigation.
  - *Wattar* represents the physical moisture equilibrium where soil is dry enough to support tractor tires without causing structural compaction or clod smearing, yet moist enough (>15–18% soil moisture) to trigger seed imbibition and germination.
  - As soon as the field reaches *Wattar*, execute 2 rounds of cross-cultivation with tillers, followed immediately by heavy planking (*Suhaga*). Planking seals surface capillary pores, trapping moisture inside the seed zone.

- **Critical Sowing Depth Metrics:**
  - *Optimal Sowing Depth:* Seeds must be placed at a precise depth of 3.0 to 5.0 cm into moist soil.
  - *Hazards of Shallow Sowing (< 2.0 cm):* Exposes seeds to rapid surface soil desiccation under hot winds, high soil temperatures, bird predation, and incomplete seed imbibition, causing germination failure.
  - *Hazards of Deep Sowing (> 6.0 cm):* Exceeds the hypocotyl elongation capacity of the embryo. Cotyledons exhaust energy reserves before reaching the surface, resulting in pre-emergence seedling mortality, delayed emergence, and weak plants.

- **Planting Modifications for Salt-Affected and Saline Soils:**
  - In soils suffering from high electrical conductivity (EC) or where irrigation water is moderately saline, soluble salts migrate upward via capillary action and concentrate on ridge crests and bed tops during surface evaporation.
  - *Salinity Mitigation Sowing Protocol:* Avoid planting on top of ridges or raised beds. Dibble seeds inside the furrows or on the lower slope of ridges (below the salt accumulation line), ensuring root zone salinity remains within tolerable limits for germinating seeds.

1.7 Gap Filling Technology and Seedling Establishment Protocols

- **Agronomic Rationale for Gap Filling:**
  - Field seedling mortality caused by soil crusting, high heat, or seed rots creates gaps along crop rows, reducing plant population density below economic thresholds.
  - Maintaining target plant stand is essential for maximizing yield per unit area. Gap filling using inherent soil moisture within 5 to 8 Days After Sowing (DAS) establishes uniform plant populations without requiring early flood irrigation.
- **Inherent Moisture Gap Filling Technology (Step-by-Step Methodology):**
  - *Step I - Field Inspection (5–8 DAS):* Inspect field rows 5 to 8 days after sowing to identify non-germinated spots or row gaps exceeding 45 cm.
  - *Step II - Seed Pre-Soaking:* Soak delinted cotton seeds in clean water for 2 to 4 hours prior to field operations to initiate internal seed hydration.
  - *Step III - Dry Soil Removal:* Scrape away the upper dry surface crust (1–2 cm) at the gap spot using a small hand trowel until the underlying moist soil layer is exposed.
  - *Step IV - Hole Creation:* Create a small planting hole (3–4 cm deep) in the moist soil layer using a handheld wooden seed dibbler or trowel.
  - *Step V - Seed Placement Covering:* Dibble 1 to 2 pre-soaked seeds into the moist hole, gently covering them with moist soil, followed by a light layer of dry soil to prevent capillary evaporation.
  - *Step VI - Accelerated Emergence:* Pre-soaked seeds germinate within 4 to 5 days, aligning canopy growth with the original crop stand. One worker can gap-fill 10% to 20% gaps across one hectare in a single day.
- **Prohibition of Early Flood Irrigation for Gap Filling:** Applying early flood irrigation before 45 DAS to force germination of gap-filled seeds is strictly discouraged. Early flooding causes surface crusting, suffocates seedling roots, restricts vertical taproot penetration, and encourages shallow lateral rooting.

1.8 Pre-Emergence Weed Suppression and Herbicide Stewardship

- **Critical Period of Crop-Weed Competition (CPC):**
  - Cotton seedlings grow slowly during the first 45 to 60 DAS, leaving open canopy spaces that encourage aggressive weed emergence.
  - Uncontrolled weed growth during this critical 60-day window causes severe nutrient, moisture, and light competition, leading to crop stunting and seed cotton yield losses ranging from 40% to 85%.
- **Major Early Season Weed Flora in Cotton Fields:**

  - *Sedges:* Purple nutsedge / Motha (*Cyperus rotundus*).
  - *Broadleaf Weeds:* Horse purslane / Itsit (*Trianthema portulacastrum*), Pigweed / Chaulai (*Amaranthus viridis*), False amaranth (*Digera arvensis*), Field bindweed (*Convolvulus arvensis*), and Morning glory (*Ipomoea* spp.).
  - *Grasses:* Bermudagrass / Doob ghas (*Cynodon dactylon*), Crowfoot grass / Gurta ghas (*Dactyloctenium aegyptium*), Barnyard grass (*Echinochloa crus-galli*), and Crabgrass (*Digitaria sanguinalis*).

- **Soil-Active Pre-Emergence Herbicide Protocols:**
  - *Pendimethalin Formulations:* Apply Pendimethalin 30% EC @ 2.5 to 3.3 L/ha (0.75–1.0 kg a.i./ha) or Pendimethalin 38.7% CS @ 1.75 to 2.0 L/ha as a soil-active uniform spray within 24 to 48 hours of sowing.
  - *Fluchloralin Formulation:* Apply Fluchloralin 45% EC @ 2.0 to 2.5 L/ha as a pre-plant soil incorporation (PPI) or pre-emergence spray on moist seedbed soil.
  - *Application Requirements:* Herbicides must be dissolved in 500 liters of water per hectare and sprayed using a tractor-mounted boom sprayer or knapsack sprayer equipped with a flat-fan nozzle under uniform pressure.

- **Intercropping Cover Crops for Biological Weed Suppression:**
  - Intercrop short-duration cover crops like cowpea, green gram, or soybean in a 1:2 row ratio between wide-spaced cotton rows.
  - Intercrops form a dense biological ground cover that suppresses weed seed germination, reduces soil erosion, fixes atmospheric nitrogen, and serves as a natural breeding hub for predatory biocontrol agents.

1.9 Early Biotic and Abiotic Risk Management during Planting

- **Soil Crusting and Emergence Mechanical Impedance:**
  - *Risk Factors:* Heavy rainstorms occurring shortly after sowing (1–4 DAS) followed by intense sunshine dry out the soil surface, forming a dense, hard crust.
  - *Emergence Failure:* Elongating hypocotyl arches strike the hard crust and break, preventing cotyledons from reaching the surface, resulting in catastrophic stand failure.
  - *Mechanical Crust-Breaking Intervention:* Operate a light wooden planker, spike-tooth harrow, blind cultivator, or rotary hoe gently across rows as soon as soil surface dries to shatter the crust without disturbing deeply placed seeds.

- **Seedling Burning and Thermal Heat Stress Mitigation:**
  - *Risk Factors:* Extreme surface soil temperatures ($> 45^{\circ}$C) in April–May scorch tender hypocotyl stems at the soil line, producing dark dry girdling rings that kill emerging seedlings.

  - *Mitigation Protocols:* Ensure heavy pre-sowing *Rauni* irrigation to charge subsoil moisture; sow seeds on the eastern or northern slope of ridges to provide shade during peak afternoon heat; deploy overhead sprinkler irrigation (if available) during extreme heat waves.

- **Prohibition of Early Chemical Sprays (0–60 DAS):**
  - Chemical insecticide sprays (especially broad-spectrum organophosphates, synthetic pyrethroids, or foliar neonicotinoids) must be strictly prohibited during the first 60 Days After Sowing.
  - Systemic insecticidal seed treatments (Imidacloprid/Thiamethoxam) provide adequate protection against early sucking pests. Avoiding foliar sprays preserves generalist predators (*Chrysoperla carnea*, coccinellids, spiders), preventing secondary pest outbreaks of whiteflies and aphids.

- **Yellow Sticky Trap Deployment at Planting:** Install yellow sticky traps @ 25 traps/ha (10 traps/acre) at canopy height immediately following crop emergence to monitor and trap adult whiteflies and jassids.

1.10 Integrated Field Planting Master Operational Summary

- **Chronological Stage-Wise Sowing Roadmap:**
  1. *Pre-Sowing Phase (April 1 to April 15):* Test soil for NPK and micronutrients; execute deep summer Mouldboard ploughing; cross-subsoil at 1.0 m spacing to shatter hardpans; level field with Laser Land Leveller; apply 5–10 t/ha FYM and 250 kg/ha Neem Cake; apply heavy pre-sowing *Rauni* irrigation with canal water.
  2. *Seed Preparation Phase (April 15 to April 30):* Acid-delint fuzzy seeds with concentrated H_2SO_4 (100 mL/kg); dress seeds in sequential order ($F \rightarrow I \rightarrow B$) using Carboxin 37.5% + Thiram 37.5% @ 3.5 g/kg, Imidacloprid 70% WS @ 7 g/kg, and *Trichoderma viride* @ 5–10 g/kg seed; soak in Streptomycin solution (40 ppm) for bacterial blight.
  3. *Field Sowing Operation Phase (April 20 to May 15):* Wait for soil to reach *Wattar* moisture; cultivate 2 times with tillers and plank with *Suhaga*; sow seeds at 3–5 cm depth using a pneumatic/inclined plate planter or manual dibbler during morning/evening hours; apply pre-emergence herbicide Pendimethalin 30% EC @ 2.5–3.3 L/ha within 48 hours.
  4. *Post-Sowing Establishment Phase (May 15 to May 30):* Inspect field 5–8 DAS; execute inherent moisture gap filling using pre-soaked seeds and hand dibblers; break soil crusts with a rotary hoe if rains occur; perform seedling thinning at 15–20 DAS to maintain target plant density; install yellow sticky traps @ 25 traps/ha; refrain from early broad-spectrum chemical sprays.

- **Core Field Planting Operational Integration Summary:**
  - *Agro-Climatic Window:* April 1 to May 15 in North Zone; June 15 to July 15 in Central Zone. Avoids seedling heat burning and early CLCuD outbreaks.

- *Density Metrics:* American Bt Hybrids (2.375–2.970 kg/ha seed) spaced at 67.5–108 cm × 45–90 cm (15,432 to 24,691 plants/ha); HDPS American Cotton spaced at 67.5 cm × 15 cm (98,765 plants/ha); Desi Varieties (7.5–15.0 kg/ha seed) spaced at 67.5 cm × 30–45 cm (32,921 to 49,382 plants/ha).
- *Seed Treatment:* Concentrated H_2SO_4 acid delinting (100 mL/kg) followed by sequential $F \rightarrow I \rightarrow B$ dressing with Carboxin + Thiram (3.5 g/kg), Imidacloprid 70% WS (7 g/kg), and *Trichoderma* (5–10 g/kg).
- *Sowing Precision:* Tractor pneumatic / inclined plate planters or manual dibbling at 3–5 cm depth into *Wattar* soil moisture; pre-emergence Pendimethalin 30% EC (@ 2.5–3.3 L/ha) within 48 hours; inherent soil moisture gap filling at 5–8 DAS.